How Semantics Interacts with Emotional Prosody in the Bilingual Brain?

Introduction. Emotional information in spoken language is conveyed through both semantic and prosodic cues. Simultaneously processing these cues can be challenging for non-native speakers, particularly in tonal languages like Mandarin Chinese, where pitch encodes both semantic and emotional prosody (Xu, 2005). Behavioral findings show that semantics shapes emotional prosody perception differently in native and non-native Chinese speakers (Xiao & Liu, 2024). However, the neural bases of these processes remain unclear in this tonal language. Therefore, this study employed functional Magnetic Resonance Imaging (fMRI) to investigate the neural mechanisms underlying the processing of semantics and emotional prosody in native and non-native Chinese speakers.

Participants and Stimuli. A total of 30 right-handed, neurologically healthy adults (18F12M, M_{age} = 22.9) were included in the study: 15 L1-Chinese L2-English speakers and 15 L1-English L2-Chinese learners. To ensure comparable semantic knowledge, all L2 learners achieved an accuracy rate of at least 80% on a pre-experiment vocabulary screening task. The critical manipulation of the stimuli was the congruence between semantics (positive, negative, neutral) and emotional prosody (positive, negative). All stimuli were controlled for syllable length and lexical tone. Positive and negative semantic stimuli were additionally matched for frequency, valence, arousal, concreteness, and age of acquisition. A female and a male native Chinese speaker recorded all 1,152 auditory stimuli with positive and negative emotional prosody, and valence and arousal ratings from 42 native speakers (23F19M, M_{age} = 22.6) confirmed that positive prosody was perceived as positive and negative prosody as negative.

fMRI Design. Two field maps and a short practice run were acquired at the start of the session to familiarize participants with the task and response device. A block design was used across four 11-minute functional runs, with each run containing 24 randomized blocks. Each block lasted approximately 26 seconds, during which participants listened to 12 auditory stimuli and then judged whether the speaker sounded like they were experiencing negative or positive emotions. A 6-minute high-resolution anatomical scan was obtained at the end of the session. The run order and finger assignments were counterbalanced across participants. MRI data were acquired on a Siemens 3T Prisma Fit scanner with a 20-channel head coil, preprocessed with fMRIPrep (Esteban et al., 2019), and analyzed using AFNI (Cox, 1996).

Results. Behaviorally (Figure 1), the native group recognized emotional prosody more accurately than the learner group (B = 1.682, p = 0.002). Both groups were more accurate in semantics-prosody congruent than incongruent conditions (B = 0.480, p < 0.001). Whole-brain analyses indicated that negative emotional prosody elicited greater activation in temporal regions, whereas positive emotional prosody activated frontal regions more strongly (Figure 2). Furthermore, when semantics and prosody were congruent, we observed greater activation in the bilateral angular gyrus, posterior cingulate cortex, and precuneus. Similar regions were also activated for the incongruent condition, with additional activation in the bilateral inferior frontal gyrus (IFG) (Figure 3). ROI analyses revealed significant group differences: compared to L2 learners, native speakers showed greater activation in the semantics-prosody interface regions, including the superior temporal gyrus (STG, for tone processing), middle temporal gyrus (MTG, for meaning and sound integration), and IFG (for evaluative functions).

Conclusion. Emotional prosody in Mandarin Chinese recruits a fronto-temporo-parietal network, consistent with findings from non-tonal languages (e.g., Seydell-Greenwald et al., 2020). These findings highlight the interplay between semantics and emotional prosody in a tonal language and also clarify the similarities and differences in neural mechanisms between native and non-native Chinese speakers.

References

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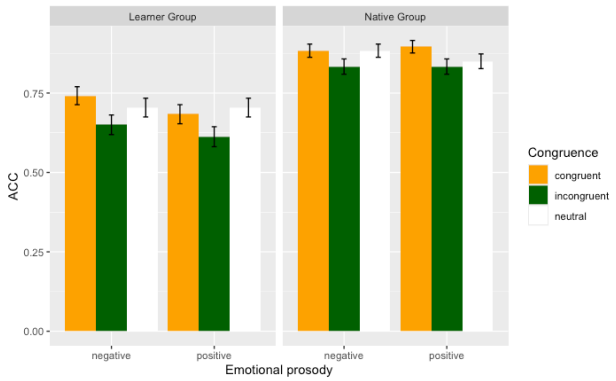

Figure 1. Mean accuracy for the native and learner groups in the semantic congruent (orange bars), incongruent (green bars), and neutral (white bars) conditions across negative and positive emotional prosody. Error bars represent 95% confidence intervals.

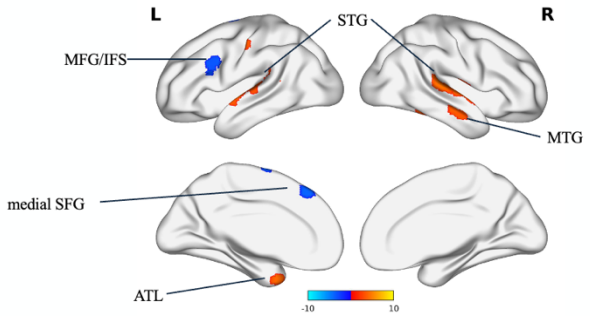

Figure 2. Whole-brain activation (z-score) for all native and non-native speakers in the contrast of negative (orange) versus positive (blue) emotional prosody. Statistical maps are projected on the MNI152 template at a voxel-wise threshold of $p < 0.001$, cluster-level corrected at $p < 0.05$.

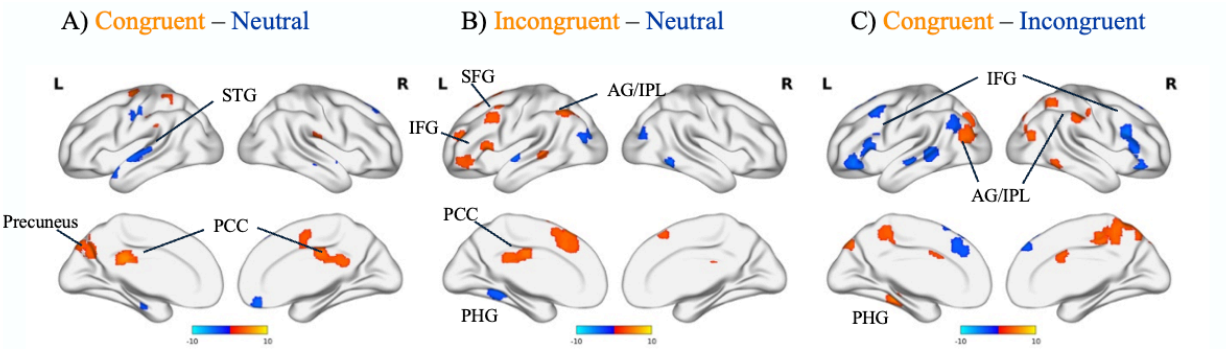

Figure 3. Whole-brain activation (z-score) for all native and non-native speakers in the contrasts A) between congruent and neutral, B) between incongruent and neutral, and C) between congruent and incongruent. Statistical maps are projected on the MNI152 template at a voxel-wise threshold of $p < 0.001$, cluster-level corrected at $p < 0.05$.